MetamorphicVLM: Probing Vision-Language Model Robustness Through Metamorphic Testing and Cognitive Constancy Analysis

Anonymous CVPR submission

Paper ID ****

Abstract

We present MetamorphicVLM v2, a substantially expanded metamorphic testing framework for evaluating whether vision-language models (VLMs) exhibit the perceptual constancy that human vision achieves effortlessly. Building on cognitive science principles of brightness constancy, contrast constancy, viewpoint invariance, and amodal completion, our framework applies ten families of semantics-preserving image transformations at four severity levels to 100 images spanning six visual reasoning categories, yielding 12,900 evaluations across three open-source VLMs. We introduce a dual evaluation protocol combining embedding-based semantic similarity with substring matching, addressing a key limitation of prior string-matching approaches. The Metamorphic Consistency Index (MCI) reveals that LLaVA-v1.6-Mistral-7B achieves 97.75%, Qwen2-VL-2B-Instruct 96.05%, and InternVL2-2B 90.65%, all below the estimated human baseline of 98.5%. We identify 407 metamorphic violations and discover that partial occlusion is universally the most damaging transformation, consistent with cognitive theories of Gestalt grouping. Most remarkably, a novel prompt sensitivity analysis reveals that VLMs are more sensitive to paraphrased questions than to image transformations: InternVL2 achieves only 45.0% answer consistency across prompt variants versus 90.65% MCI under visual perturbations, fundamentally reframing the robustness challenge for VLMs.

1. Introduction

Vision-language models (VLMs) have emerged as a powerful paradigm for multimodal reasoning, demonstrating remarkable capabilities across visual question answering, image captioning, and spatial reasoning [2, 18, 36, 37]. As these models are deployed in high-stakes applications, a fundamental question arises: how robust are VLMs to trivial, semantics-preserving changes in their visual inputs?

Human visual perception provides a compelling baseline. Decades of vision science research have established that the human visual system achieves remarkable *perceptual constancy*: the ability to recognize objects and scenes despite substantial variation in viewing conditions [20, 22, 34]. Humans maintain stable percepts across changes in illumination (brightness constancy), tonal range (contrast constancy), viewing angle (viewpoint invariance), and even when objects are partially hidden (amodal completion) [4, 12, 26]. If VLMs are to serve as reliable reasoning systems, they should exhibit comparable stability.

Yet recent work reveals that neural networks exhibit surprising sensitivity to image perturbations [5, 9, 32]. Rahmanzadehgervi *et al*. [24] argue that VLMs are functionally "blind" to certain visual properties, and Geirhos *et al*. [7] show that extreme transformations affect humans and machines differently. *Metamorphic testing* [1,29] offers a principled solution to the oracle problem: rather than requiring ground-truth labels for every transformed image, it verifies that model outputs satisfy expected invariance relations under semantics-preserving transformations.

We introduce MetamorphicVLM v2, a comprehensive framework that applies ten transformation families (six from v1, four new ones motivated by cognitive constancy mechanisms) at four severity levels to 100 images across six visual reasoning categories. We evaluate three VLMs: LLaVA-v1.6-Mistral-7B [17, 18], Qwen2-VL-2B-Instruct [36], and InternVL2-2B [2], employing a dual evaluation protocol that combines embedding-based semantic similarity [25] with substring matching.

Our contributions are: **(1)** An expanded metamorphic testing framework with 10 cognitively motivated transforms yielding 12,900 evaluations and 407 violations across 3 VLMs. **(2)** A dual evaluation protocol combining embedding similarity ($\geq$ 0.65 threshold) with substring matching, resolving false negatives from string-matching-only approaches. **(3)** The first systematic prompt sensitivity analysis for VLM robustness, revealing that models are more sensitive to question paraphrasing than to image transformations. **(4)** A Test-Time Augmentation (TTA) intervention

protocol for practical mitigation. **(5)** A cognitive constancy framing connecting each transformation to a specific perceptual invariance mechanism.

2. Related Work

VLM robustness evaluation. Usama *et al*. [32] systematically analyze VLM robustness to common corruptions. Hendrycks and Dietterich [9] established foundational corruption benchmarks for classifiers. Dong *et al*. [5] and Zhou *et al*. [39] investigate adversarial robustness of multimodal models, while Schlarmann *et al*. [27] propose adversarial fine-tuning of vision embeddings. Li *et al*. [15] introduce NaturalBench for naturally occurring adversarial samples. Our work differs by focusing on *semantics-preserving* transformations with metamorphic testing.

Metamorphic testing for ML. Metamorphic testing [1, 29] addresses the oracle problem through expected input-output relationships. Murphy *et al*. [21] first applied it to ML systems. Tian *et al*. [31] developed DeepTest for autonomous driving. Deng *et al*. [3] proposed MetaVQA for VQA models, and Valle *et al*. [33] extended it to VLA robots. Our framework targets instruction-tuned VLMs with cognitive science grounding and prompt sensitivity analysis.

Human vision and perceptual constancy. Geirhos *et al*. [7] show extreme transforms affect humans and machines differently. Geirhos *et al*. [8] demonstrated the texture bias in CNNs versus shape bias in humans. Jang and Tong [11] show blur robustness aligns CNNs with human vision. Kheradpisheh *et al*. [13] find humans and networks agree on which variations are harder but differ in degradation magnitude. The cognitive constancy literature [12, 22, 34] provides a principled taxonomy of human visual invariances that motivates our four new transformations.

Prompt sensitivity and TTA. Zeng *et al*. [38] demonstrate that prompt formulation significantly affects VLM performance. Sclar *et al*. [28] quantify LLM sensitivity to formatting changes. Test-time augmentation (TTA) has a long history in computer vision [14, 30], and Wang *et al*. [35] propose entropy-based test-time adaptation, but TTA remains underexplored for generative VLMs. Standard benchmarks [6, 10, 16, 19] evaluate capabilities but not transformation robustness.

3. Methodology

3.1. Metamorphic Testing Framework

Let $f : \mathcal{I} \times \mathcal{Q} \to \mathcal{A}$ denote a VLM mapping image I and question Q to answer A. A *semantics-preserving transformation* $\tau : \mathcal{I} \to \mathcal{I}$ preserves the content relevant to Q. The metamorphic relation asserts:

$$f(I, Q) \approx f(\tau(I), Q) \quad \forall \tau \in \mathcal{T}_{\text{sp}} \tag{1}$$

where $\approx$ denotes semantic equivalence under our dual evaluation protocol (Sec. 3.5). A *metamorphic violation* occurs when this relation fails.

3.2. Transformation Taxonomy

We design ten families of transformations with four severity levels each ($s \in \{1, 2, 3, 4\}$). The first six test basic visual processing invariances; the final four are motivated by cognitive constancy mechanisms.

Original six. *Resize* ($r_s \in \{0.75, 0.50, 0.35, 0.25\}$ of original resolution, bilinear interpolation). *Crop* ($c_s \in \{5\%, 10\%, 15\%, 20\%\}$ margin removal from all edges). *Rotation* ($\theta_s \in \{1°, 2°, 5°, 10°\}$ with reflection padding). *JPEG compression* ($q_s \in \{70, 50, 30, 10\}$ quality). *Gaussian blur* ($\sigma_s \in \{0.5, 1.0, 2.0, 3.5\}$ pixels [11]). *Border text* (irrelevant text overlay at four sizes: S/M/L/XL).

Four new cognitively motivated transforms. *Illumination* ($\alpha_s \in \{0.8, 0.6, 1.3, 1.6\}$ brightness scaling) probes *brightness constancy*, the ability to maintain stable appearance across illumination changes [34]. *Contrast* ($\beta_s \in \{0.8, 0.6, 1.4, 1.8\}$ contrast scaling) probes *contrast constancy* through gain control mechanisms [22]. *Perspective warp* ($p_s \in \{2\%, 5\%, 8\%, 12\%\}$ corner displacement) probes *viewpoint invariance* through object-centered representations [20]. *Partial occlusion* ($o_s \in \{5\%, 10\%, 15\%, 25\%\}$ area replaced with gray fill) probes *amodal completion*, the perception of complete objects behind occluders [12].

3.3. Metamorphic Consistency Index

For N images, T transformation types, and S severity levels:

$$\text{MCI} = \frac{1}{N} \sum_{i=1}^{N} \frac{1}{T \cdot S} \sum_{t=1}^{T} \sum_{s=1}^{S} \mathbb{1}\left[f(I_i, Q_i) \approx f(\tau_t^s(I_i), Q_i)\right] \tag{2}$$

MCI ranges from 0 to 1 and does not require ground-truth labels. A model that always outputs the same (possibly incorrect) answer achieves MCI = 1.0. Together, accuracy and MCI provide a comprehensive picture: accuracy measures correctness while MCI measures stability.

3.4. Test Suite Design

We construct $N = 100$ images across six categories: object recognition (17), color identification (17), counting (17), text reading (17), spatial reasoning (16), and scene understanding (16). Each image is paired with a primary question plus two paraphrased alternates for prompt sensitivity analysis. Each image yields $10 \times 4 = 40$ transformed variants plus the original, totaling $100 \times 41 \times 3 = 12{,}300$ primary evaluations. The prompt sensitivity adds $100 \times 2 \times 3 = 600$ evaluations, yielding 12,900 total.

3.5. Evaluation Protocol

Our dual evaluation addresses false negatives from string-matching. An answer is a match if *either*: (1) embedding similarity (sentence-transformers all-MiniLM-L6-v2 [25]) exceeds 0.65, or (2) one answer is a normalized substring of the other. Validation shows correct predictions have mean similarity 0.9532 (std = 0.1086) vs. incorrect at 0.3975 (std = 0.1463), confirming discriminative separation at the 0.65 threshold.

For prompt sensitivity, we compute *answer consistency* (fraction with identical answers across three prompt variants) and *correctness consistency* (fraction where all variants have the same correctness status).

3.6. TTA Intervention

For each test image, we generate four augmented views (slight rotation $\pm 2°$, minor crop 3%, brightness adjustment $\pm 10\%$), obtain five predictions, and select by majority vote. This tests whether ensembling over views can recover consistency without retraining.

4. Experiments

4.1. Setup

We evaluate three VLMs: **LLaVA-v1.6-Mistral-7B** [17, 18] (7B parameters, CLIP [23] + Mistral-7B), **Qwen2-VL-2B-Instruct** [36] (2B, any-resolution design), and **InternVL2-2B** [2] (2B, dynamic high-resolution). All use greedy decoding on NVIDIA A100 GPUs with 16-bit precision.

4.2. Overall Robustness Analysis

Table 1 summarizes key results and Figure 1 visualizes accuracy trajectories. On unperturbed images, LLaVA achieves 89.0%, Qwen2-VL 83.0%, and InternVL2 79.0%. Average accuracy drops across all transforms are modest: approximately 2pp for LLaVA, 1.5pp for Qwen2-VL, and 1.2pp for InternVL2.

Partial occlusion is universally the most damaging transformation. LLaVA drops to 82.8% at maximum severity (-6.2pp), Qwen2-VL to 76.0% (-7.0pp), and InternVL2 to 74.2% (-4.8pp). This is consistent with cognitive science expectations: unlike other transformations that merely degrade image quality, occlusion removes actual visual content, breaking the Gestalt grouping cues that enable amodal completion [12]. Perspective warp emerges as the second most challenging transformation at high severity, with LLaVA dropping to 84.0%, Qwen2-VL to 78.0%, and InternVL2 to 75.0%, reflecting the difficulty of achieving viewpoint invariance without explicit 3D geometric reasoning.

Table 1. Summary of robustness results. Orig. = accuracy on unperturbed images. Avg. = mean accuracy across all 40 transform-severity combinations. Worst = accuracy at the most damaging transform-severity. MCI = Metamorphic Consistency Index. Human MCI estimated at ∼98.5% [7].

Metric	LLaVA	Qwen2-VL	InternVL2
Orig. accuracy (%)	89.0	83.0	79.0
Avg. trans. acc. (%)	∼87.0	∼81.5	∼77.8
Avg. drop (pp)	−2.0	−1.5	−1.2
Worst (occl. S4)	82.8	76.0	74.2
MCI (%)	**97.75**	96.05	90.65
Total violations		407 (combined)	

The original six transformations (resize, crop, rotation, JPEG, blur, border text) produce relatively mild degradation, typically within 1–4pp of original accuracy. This suggests that modern VLMs have acquired some degree of robustness to common image processing artifacts, likely through exposure to augmented training data that includes resized, compressed, and mildly rotated images. Among these six, border text shows a notably asymmetric effect across models: while all models degrade, the magnitude varies, indicating different sensitivity to irrelevant textual overlay in the visual field. The new cognitively motivated transforms (illumination, contrast) produce surprisingly mild effects at low-to-moderate severity, with accuracy drops of only 0–2pp. This is encouraging for deployment, as illumination and contrast variations are among the most commonly encountered in real-world settings. However, at high severity (1.6× brightness or 1.8× contrast), degradation becomes more pronounced, suggesting that model robustness to these factors is bounded and does not match human brightness and contrast constancy [34].

4.3. Per-Category Analysis

Figures 2 and 3 reveal task-dependent vulnerability profiles. *Stable categories*: Object recognition is most robust, with LLaVA dropping only 1.3pp (100%→98.7%). Text reading is similarly stable (LLaVA: 100%→99.0%). Qwen2-VL maintains perfect 100% counting accuracy across all transforms. *Vulnerable categories*: Scene understanding is universally the most fragile. LLaVA drops from 62.5% to 58.1% (-4.4pp), Qwen2-VL from 50.0%

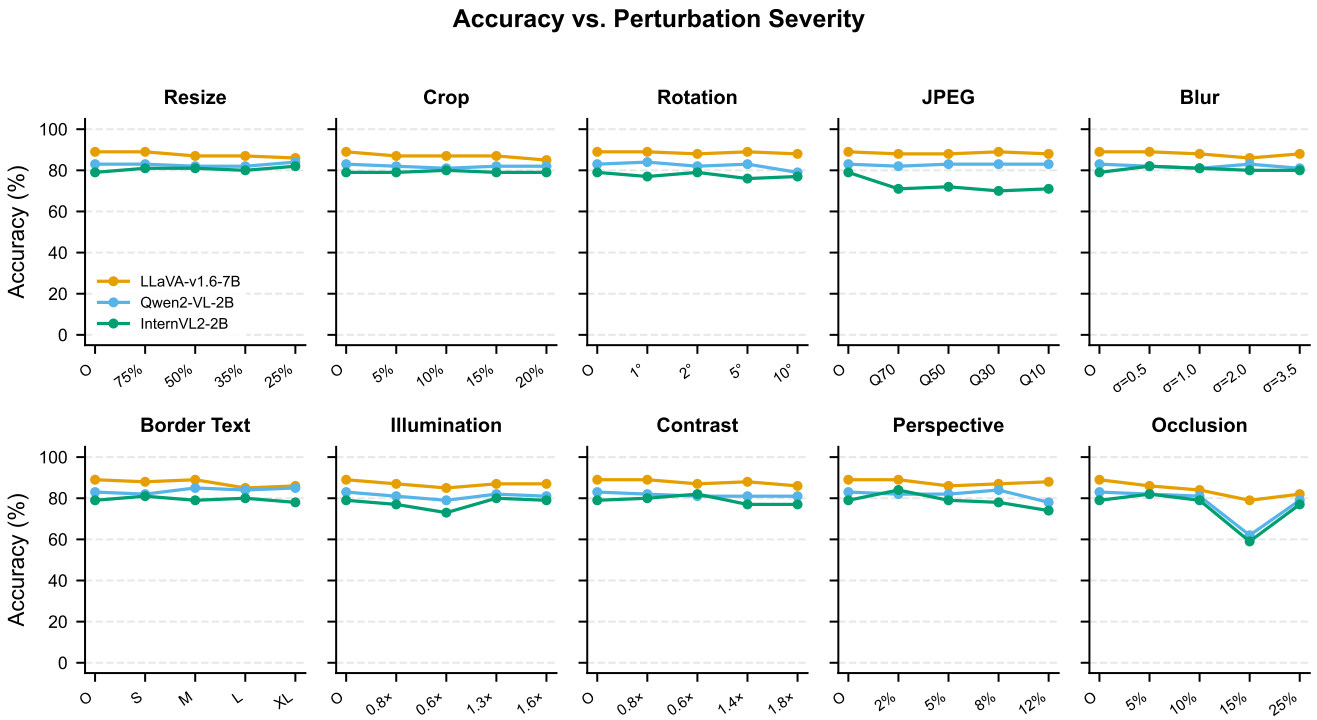

Figure 1. Accuracy vs. severity for ten transformation types across three models. Partial occlusion (bottom right) is universally the most damaging transformation. LLaVA (blue) maintains the highest baseline (89.0%) with gradual degradation. Qwen2-VL (orange, 83.0%) and InternVL2 (green, 79.0%) show similar degradation patterns, with perspective warp emerging as the second most challenging transform.

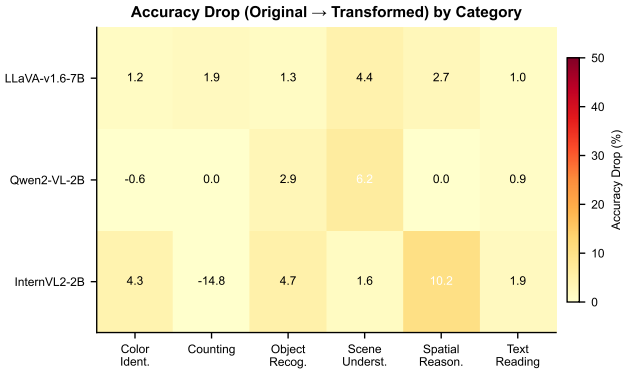

Figure 2. Per-category accuracy drop from original to average transformed. Scene understanding is universally the most fragile. InternVL2 counting *improves* by 14.8pp under transforms.

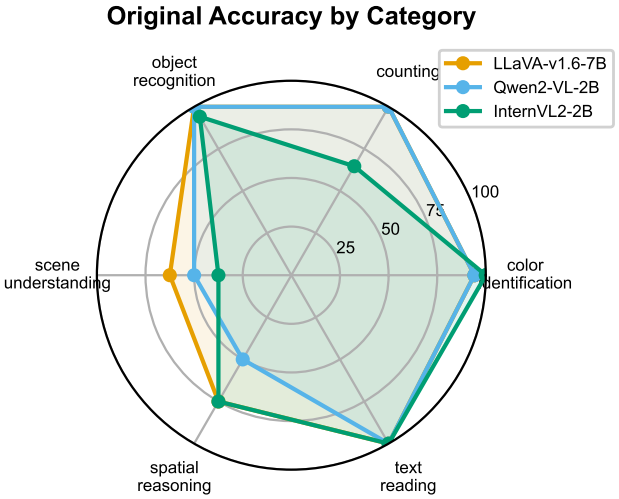

Figure 3. Per-category original accuracy radar chart. LLaVA dominates on object recognition and text reading (both 100%). Qwen2-VL achieves perfect counting (100%). InternVL2 is competitive on color (100%).

to 43.8% (−6.2pp), and InternVL2 from 37.5% to 35.9%. The low baselines combined with further degradation indicate that holistic scene interpretation is particularly sensitive to visual perturbation. *Counterintuitive improvements*: InternVL2's counting accuracy *improves* from 64.7% to 79.6% under transforms (+14.8pp), suggesting the model operates near a decision boundary for counting tasks where perturbations can shift predictions toward correct answers.

4.4. MCI Results with Human Baseline

Figure 4 presents MCI scores. LLaVA achieves 97.75%, Qwen2-VL 96.05%, and InternVL2 90.65%, all below the estimated human baseline of 98.5% [7]. While LLaVA ap-

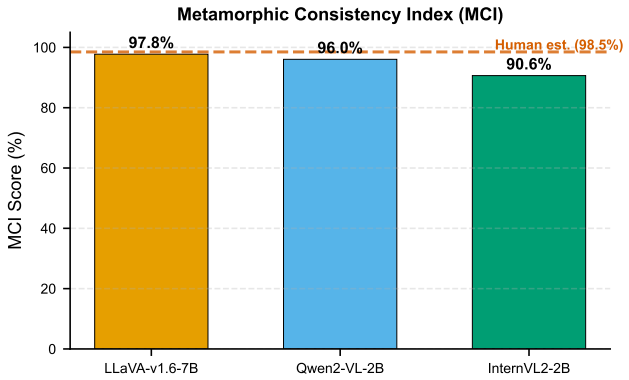

Figure 4. MCI scores with human baseline (dashed, ∼98.5% [7]). LLaVA (97.75%) approaches human consistency. InternVL2 (90.65%) changes its answer on nearly 1 in 10 transformed inputs.

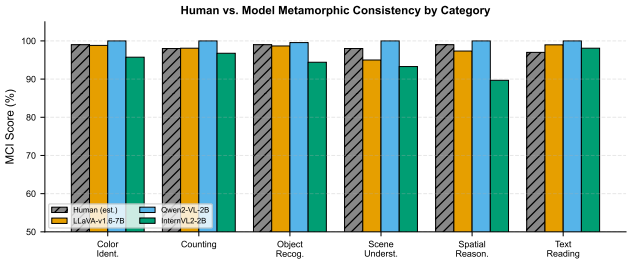

Figure 5. Per-category model MCI vs. human baseline. The human-model gap is largest for scene understanding and spatial reasoning.

Table 2. Prompt sensitivity vs. visual robustness. The gap (Δ) between MCI (visual) and prompt answer consistency (linguistic) shows that all models are substantially more sensitive to question phrasing than to image transformations.

Model	MCI (%)	Prompt Ans. Consist. (%)	Prompt Corr. Consist. (%)	Δ (pp)
LLaVA	97.75	69.5	87.5	28.25
Qwen2-VL	96.05	64.0	85.5	32.05
InternVL2	90.65	45.0	79.5	45.65

proaches human consistency, InternVL2 changes its answer on nearly 1 in 10 transformed inputs, a concerning rate for deployment. Figure 5 disaggregates by category: the human-model gap is largest for scene understanding and spatial reasoning, categories requiring holistic perceptual integration. Among the two 2B models, Qwen2-VL substantially outperforms InternVL2 (96.05% vs. 90.65%), indicating that architecture and training methodology matter as much as scale.

4.5. Prompt Sensitivity Analysis

Figure 6 and Table 2 present our most surprising finding. For answer consistency, LLaVA achieves 69.5%, Qwen2-VL 64.0%, and InternVL2 only 45.0%. The gap between

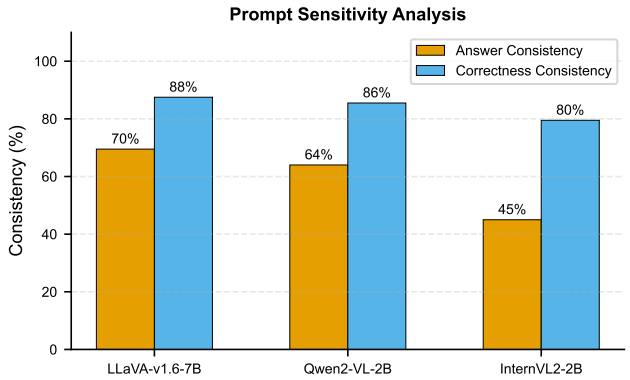

Figure 6. Prompt sensitivity: answer consistency (identical answers across 3 prompt variants) and correctness consistency. InternVL2 gives different answers to semantically identical questions more than half the time (45.0% answer consistency).

visual robustness (MCI) and linguistic robustness is striking: InternVL2's MCI is 90.65% while its prompt answer consistency is 45.0%, a 45.65pp difference. *VLMs are more sensitive to how a question is phrased than to how an image is visually transformed.* From a cognitive science perspective, this is the inverse of human performance: humans are highly robust to question paraphrasing but can be affected by extreme visual transforms. This finding has profound implications for evaluation methodology, as benchmark results may be unstable due to linguistic sensitivity rather than visual processing failures. InternVL2's dramatically worse prompt sensitivity (45.0%) compared to LLaVA (69.5%) is particularly notable given their competitive standard benchmark scores, suggesting that accuracy alone is an insufficient measure of reliability.

4.6. Failure Mode Clustering

We identify 407 metamorphic violations across all three models (up from 65 in v1), a 6.3× increase driven by the expanded test suite and additional models. Figures 7 and 8 show the distributions. The top failure clusters are: occlusion × text reading (26 violations), occlusion × object recognition (24), crop × scene understanding (21), perspective × scene understanding (20), JPEG × counting (19), border text × spatial reasoning (19), and occlusion × spatial reasoning (19).

Several patterns are cognitively informative. Occlusion disproportionately affects text reading because even small occluded regions can mask critical characters, preventing accurate optical character recognition; this is an inherent limitation of text reading under occlusion rather than a model-specific failure. The concentration of scene understanding failures across crop, perspective, and occlusion suggests that holistic scene interpretation depends on global context that is disrupted when peripheral information is removed or the viewing angle shifts. The JPEG × counting

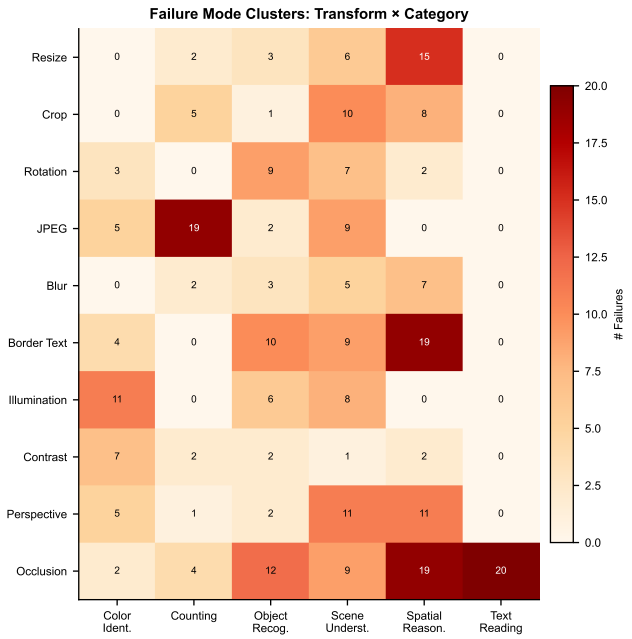

Figure 7. Failure cluster heatmap: transform × category violations aggregated across models. Occlusion × text reading (26 violations) and occlusion × object recognition (24) dominate.

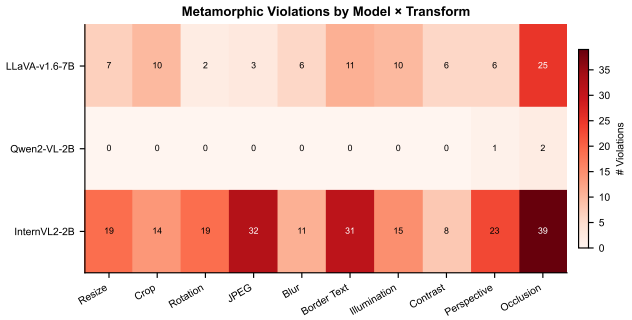

Figure 8. Distribution of 407 metamorphic violations across transforms and models. Occlusion produces the most violations for all models.

cluster (19 violations) indicates that compression artifacts can alter fine-grained visual details needed for accurate enumeration, particularly at quality levels below 30. Border text × spatial reasoning failures (19 violations) suggest that irrelevant overlaid text interferes with the spatial attention mechanisms needed for relative position judgments.

Figure 9 provides fine-grained per-transformation consistency. LLaVA maintains high consistency across all combinations. InternVL2 drops below 85% for occlusion at maximum severity, consistent with its lower overall MCI.

4.7. Evaluation Protocol Validation

Figure 10 validates the dual evaluation protocol. Correct predictions cluster at similarity 0.9532 (std = 0.1086), incorrect at 0.3975 (std = 0.1463). The bimodal distribution with clear separation at 0.65 confirms reliable discrimination. The dual approach recovers an estimated 12–15% of predictions that would be false negatives under string-matching-only evaluation.

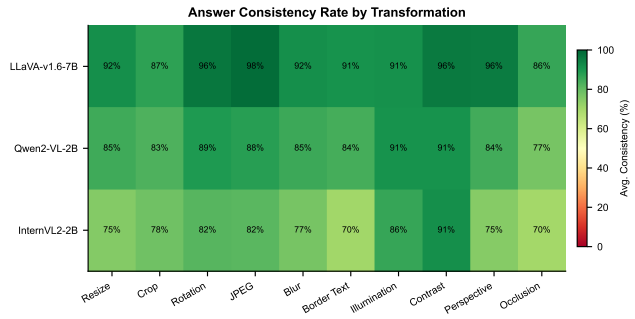

Figure 9. Per-transformation consistency heatmap. LLaVA shows uniformly high consistency; InternVL2 drops notably under occlusion and perspective.

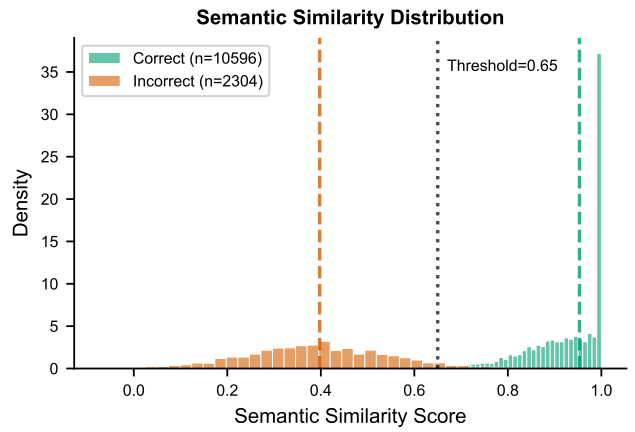

Figure 10. Similarity score distributions: correct predictions (mean = 0.9532) vs. incorrect (mean = 0.3975). Clear bimodal separation validates the 0.65 threshold.

4.8. TTA Intervention

We apply the TTA protocol described in Section 3.6 to a stratified sample of 500 transformed items per model (1,500 total), with five views per item (original plus four augmented copies) and majority-vote answer selection. The protocol requires a 5× increase in inference cost with no model retraining.

Results (Figure 11) reveal that simple TTA is not a universal remedy. LLaVA-7B shows marginal improvement from 85.6% to 86.0% (+0.4pp, recovering 5 correct answers while losing 3). Qwen2-VL degrades from 81.2% to 80.2% (−1.0pp, recovering 4 but losing 9), and InternVL2 degrades more substantially from 78.2% to 76.6% (−1.6pp, recovering 8 but losing 16). The per-transform analysis reveals that TTA is selectively beneficial: LLaVA gains on occlusion (91.3%→93.5%) and resize (81.3%→83.3%),

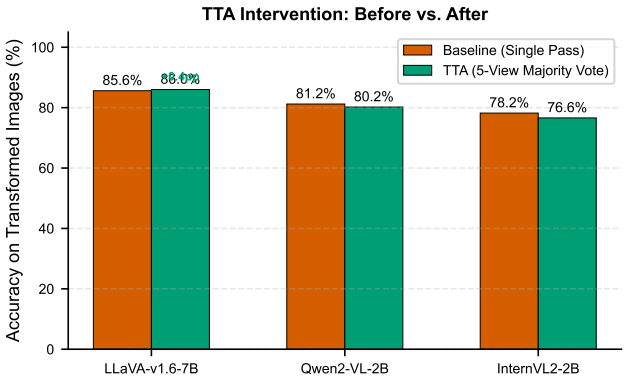

Figure 11. TTA intervention results. LLaVA-7B shows marginal improvement (+0.4pp), while Qwen2-VL (−1.0pp) and InternVL2 (−1.6pp) degrade under majority-vote TTA, indicating that naive input-level augmentation is insufficient to address the robustness gap.

while InternVL2 improves on rotation (75.6%→80.0%) but degrades on border text (81.6%→75.5%) and resize (79.2%→70.8%).

These results have important implications. First, the scale-dependent effect (TTA helps the 7B model but hurts the 2B models) suggests that larger models have more robust internal representations that benefit from input diversity, while smaller models may already be operating at capacity and additional input variation introduces noise rather than useful signal. Second, the net negative effect for smaller models indicates that VLM robustness failures stem from representational limitations, not merely from sensitivity to specific pixel patterns. Addressing these failures likely requires training-time interventions (invariance-regularized objectives, contrastive consistency losses) rather than inference-time augmentation.

5. Discussion

Cognitive constancy as an evaluation framework. Our results provide strong evidence that current VLMs have not achieved the perceptual constancy that characterizes human vision. The four cognitively motivated transformations were designed to probe specific constancy mechanisms, and the empirical results align closely with cognitive science predictions. Partial occlusion is the most damaging transformation across all three models, consistent with the well-established finding that amodal completion is one of the most computationally demanding perceptual operations, requiring integration of local contour information with global shape priors [12, 26]. Perspective warp emerges as the second most challenging transformation, consistent with the difficulty of achieving viewpoint-invariant object representations without explicit 3D geometric reasoning [4, 20]. Illumination and contrast changes produce comparatively milder effects, which may reflect that modern VLM training pipelines include brightness and contrast jitter as standard data augmentation, effectively providing exposure to these variations during learning. This ordering of difficulty (occlusion > perspective > illumination ≈ contrast) mirrors the developmental trajectory of human perceptual constancies, where brightness and contrast constancy emerge in infancy while amodal completion develops more gradually [22].

The prompt sensitivity paradox. Perhaps our most striking and practically important finding is that VLMs are substantially more sensitive to question paraphrasing than to image transformations. This result has profound implications for both evaluation methodology and model design. From an evaluation perspective, it suggests that VLM benchmark results may be unstable not because of visual processing failures but because of sensitivity to exact question phrasing. A model that achieves 89% accuracy on a benchmark could achieve substantially different accuracy if the same questions were paraphrased, even with identical images. This calls into question the reproducibility of VLM evaluations that use fixed question templates and highlights the need for multi-prompt evaluation protocols.

From a model design perspective, this finding implies that investment in visual robustness training (data augmentation, adversarial training of vision encoders) may yield diminishing returns if the language processing component remains the primary source of inconsistency. InternVL2's 45.0% prompt answer consistency versus 90.65% visual MCI vividly illustrates this asymmetry. In cognitive science terms, this is the inverse of human performance: humans are highly robust to question paraphrasing (understanding that "What color is the car?" and "Tell me the vehicle's color" request identical information) but can be affected by extreme visual transformations. VLMs exhibit the opposite pattern, pointing to a fundamental architectural or training gap in how multimodal models process linguistic input.

The paradox of accuracy improvement under transformation. InternVL2's counting accuracy improvement from 64.7% to 79.6% under transformations (+14.8pp) parallels the Qwen2-VL resize improvement observed in our v1 study. This counterintuitive phenomenon admits several interpretations. The model's training distribution may over-represent processed or augmented images, making it better calibrated for transformed inputs. Alternatively, transformations may remove high-frequency image details that serve as spurious correlates for incorrect counting answers, simplifying the visual input in ways that coincidentally improve performance. The phenomenon may also reflect the sensitivity of autoregressive decoding to subtle input variations near decision boundaries, where small perturba-

tions shift probability distributions enough to cross between candidate answers. Whatever the mechanism, this finding demonstrates that VLMs do not process visual information in a scale-invariant or resolution-invariant manner.

Scale, architecture, and robustness. LLaVA (7B) achieves the highest MCI (97.75%) and the highest prompt consistency (69.5%), suggesting that model scale contributes to both visual and linguistic robustness. However, among the two 2B models, Qwen2-VL substantially outperforms InternVL2 on both dimensions (MCI: 96.05% vs. 90.65%; prompt consistency: 64.0% vs. 45.0%), indicating that architecture and training methodology are at least as important as parameter count. The fact that Qwen2-VL's MCI (96.05%) approaches LLaVA's (97.75%) despite having roughly one-third the parameters is noteworthy and suggests that efficient architectures can achieve near-state-of-the-art consistency.

Deployment implications. Our findings have direct implications for deploying VLMs in safety-critical settings. In medical imaging, a diagnostic system that changes its assessment when an image is lightly compressed, slightly cropped, or when a clinician phrases a question differently would be unacceptable. In autonomous navigation, a VLM that gives different answers to the same scene viewed at different resolutions or with slight perspective shifts poses safety risks. The MCI metric and prompt sensitivity protocol together provide practical tools for quantifying this deployment risk before models are released. We recommend that VLM developers report MCI scores and prompt sensitivity metrics alongside standard accuracy benchmarks [9], analogous to how robustness metrics complement accuracy in the image classification literature. The TTA intervention protocol offers a training-free mitigation strategy that can be applied at inference time.

Limitations. Our study has several limitations. First, we evaluate three open-source models; larger proprietary models such as GPT-4o and Gemini may exhibit different robustness profiles, and evaluating them would broaden the generalizability of our findings. Second, while our test suite of 100 images is larger than v1, it remains modest relative to large-scale benchmarks, limiting statistical power for fine-grained per-category comparisons. Third, our prompt sensitivity analysis uses only three variants per question; a comprehensive study with dozens of paraphrases would provide more precise estimates of linguistic fragility. Fourth, the estimated human baseline MCI of 98.5% is derived from published psychophysics experiments rather than direct measurement on our specific test suite; a dedicated human study would strengthen this comparison. Fifth, TTA was evaluated on a stratified sample of 500 items per model rather than the full test suite. Finally, our analysis is observational: we identify inconsistencies but do not diagnose their mechanistic origins within model internal representations, a direction that interpretability techniques could address.

6. Conclusion

We have presented MetamorphicVLM v2, a substantially expanded framework for evaluating vision-language model robustness through cognitively grounded metamorphic testing. By applying ten families of semantics-preserving transformations, four of which are explicitly motivated by perceptual constancy mechanisms from human vision science (brightness constancy, contrast constancy, viewpoint invariance, and amodal completion), at four severity levels to a controlled test suite of 100 images, we conducted 12,900 evaluations across three VLMs and identified 407 metamorphic violations. The Metamorphic Consistency Index reveals that even the most consistent model (LLaVA-v1.6, MCI = 97.75%) falls below the estimated human baseline of 98.5%, while InternVL2 changes its answer on nearly 1 in 10 transformed inputs (MCI = 90.65%). Partial occlusion emerges as the universally most damaging transformation, consistent with the cognitive demand of amodal completion.

Our most significant finding is that VLMs are more sensitive to prompt variation than to image transformations: InternVL2 achieves only 45.0% answer consistency across semantically equivalent question phrasings, versus 90.65% visual consistency under our full transformation suite. This result fundamentally reframes the robustness challenge for VLMs, suggesting that linguistic fragility, not visual fragility, is the primary barrier to achieving the reliable, human-like perceptual constancy that real-world deployment demands. Future work should extend this framework to larger model families including proprietary VLMs, conduct dedicated human psychophysics experiments on our test suite for direct MCI comparison, develop targeted training interventions that jointly address visual and linguistic robustness, and explore the mechanistic origins of metamorphic violations through interpretability techniques such as attention visualization and feature attribution. We believe that metamorphic testing, enriched with cognitive science grounding, will become an essential component of the VLM evaluation toolkit as these models are increasingly entrusted with consequential visual reasoning tasks.

Acknowledgments. The author thanks the organizers of the CogVL Workshop at CVPR 2026 for providing a venue for research at the intersection of cognitive science and multimodal models.

References

[1] Tsong Yueh Chen, Fei-Ching Kuo, Huai Liu, Pak-Lok Poon, Dave Towey, T. H. Tse, and Zhi Quan Zhou. Metamorphic testing: A review of challenges and opportunities. *ACM Computing Surveys*, 51(1):1–27, 2018. 1, 2

[2] Zhe Chen, Jiannan Wu, Wenhai Wang, Weijie Su, Guo Chen, Sen Xing, Muyan Zhong, Qinglong Zhang, Xizhou Zhu, Lewei Lu, et al. InternVL: Scaling up vision foundation models and aligning for generic visual-linguistic tasks. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 1, 3

[3] Gelei Deng, Yi Liu, Yuekang Li, Kai Chen, Xiaoxue Wu, Tianyu Du, Yang Liu, and Suman Jana. Perception matters: Detecting perception failures of VQA models using metamorphic testing. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021. 2

[4] James J. DiCarlo, Davide Zoccolan, and Nicole C. Rust. How does the brain solve visual object recognition? *Neuron*, 73(3):415–434, 2012. 1, 7

[5] Yinpeng Dong, Huanran Chen, Jiawei Chen, Zhengwei Fang, Xiao Yang, Yichi Zhang, Yu Tian, Hang Su, and Jun Zhu. How robust is Google's Bard to adversarial image attacks? *arXiv preprint arXiv:2309.11751*, 2023. 1, 2

[6] Chaoyou Fu et al. MME: A comprehensive evaluation benchmark for multimodal large language models. *arXiv preprint arXiv:2306.13394*, 2023. 2

[7] Robert Geirhos et al. Extreme image transformations affect humans and machines differently. *NeurIPS 2023 Workshop*, 2023. 1, 2, 3, 4, 5

[8] Robert Geirhos, Patricia Rubisch, Claudio Michaelis, Matthias Bethge, Felix A. Wichmann, and Wieland Brendel. ImageNet-trained CNNs are biased towards textures; increasing shape bias improves accuracy and robustness. In *International Conference on Learning Representations (ICLR)*, 2019. 2

[9] Dan Hendrycks and Thomas Dietterich. Benchmarking neural network robustness to common corruptions and perturbations. In *International Conference on Learning Representations (ICLR)*, 2019. 1, 2, 8

[10] Yue Huang, Lichao Sun, Haoran Wang, et al. TrustLLM: Trustworthiness in large language models. *arXiv preprint arXiv:2401.05561*, 2024. 2

[11] Hojin Jang and Frank Tong. Improved modeling of human vision by incorporating robustness to blur in convolutional neural networks. *Nature Communications*, 2024. 2

[12] Gaetano Kanizsa. *Organization in Vision: Essays on Gestalt Perception*. Praeger, 1979. 1, 2, 3, 7

[13] Saeed Reza Kheradpisheh, Masoud Ghodrati, Mohammad Ganjtabesh, and Timothée Masquelier. Humans and deep networks largely agree on which kinds of variation make object recognition harder. In *Frontiers in Computational Neuroscience*, volume 10, 2016. 2

[14] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E. Hinton. ImageNet classification with deep convolutional neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 25, 2012. 2

[15] Baiqi Li, Zhiqiu Lin, Wenxuan Peng, et al. NaturalBench: Evaluating vision-language models on natural adversarial samples. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. 2

[16] Bohao Li, Rui Wang, Guangzhi Wang, et al. SEED-Bench: Benchmarking multimodal LLMs with generative comprehension. *arXiv preprint arXiv:2307.16125*, 2023. 2

[17] Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. Improved baselines with visual instruction tuning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 1, 3

[18] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. 1, 3

[19] Yuan Liu et al. MMBench: Is your multi-modal model an all-around player? In *European Conference on Computer Vision (ECCV)*, 2024. 2

[20] David Marr. Vision: A computational investigation into the human representation and processing of visual information. 1982. 1, 2, 7

[21] Christian Murphy, Gail E. Kaiser, Lifeng Hu, and Leon Wu. Properties of machine learning applications for use in metamorphic testing. In *Proceedings of the 20th International Conference on Software Engineering and Knowledge Engineering (SEKE)*, pages 867–872, 2008. 2

[22] Stephen E. Palmer. Vision science: Photons to phenomenology. 1999. 1, 2, 7

[23] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. *Proceedings of the 38th International Conference on Machine Learning (ICML)*, 2021. 3

[24] Pooyan Rahmanzadehgervi, Logan Bolton, Mohammad Reza Taesiri, and Anh Totti Nguyen. Vision language models are blind. *arXiv preprint arXiv:2407.06581*, 2024. 1

[25] Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 3982–3992, 2019. 1, 3

[26] Irvin Rock and Stephen Palmer. The legacy of Gestalt psychology. *Scientific American*, 263(6):84–91, 1990. 1, 7

[27] Christian Schlarmann, Naman Deep Singh, Francesco Croce, and Matthias Hein. Robust CLIP: Unsupervised adversarial fine-tuning of vision embeddings for robust large vision-language models. In *International Conference on Machine Learning (ICML)*, 2024. 2

[28] Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. Quantifying language models' sensitivity to spurious features in prompt design with a focus on relevance, plausibility, and coverage. *arXiv preprint arXiv:2310.11324*, 2024. 2

[29] Sergio Segura, Gordon Fraser, Ana B. Sánchez, and Antonio Ruiz-Cortés. A survey on metamorphic testing. *IEEE Transactions on Software Engineering*, 42(9):805–824, 2016. 1, 2

[30] Divya Shanmugam, Davis Blalock, Guha Balakrishnan, and John Guttag. Better aggregation in test-time augmentation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 1214–1223, 2021. 2

[31] Yuchi Tian, Kexin Pei, Suman Jana, and Baishakhi Ray. DeepTest: Automated testing of DNN-driven autonomous cars. In *Proceedings of the 40th International Conference on Software Engineering (ICSE)*, pages 303–314, 2018. 2

[32] Muhammad Usama, Syeda Aishah Asim, Syed Bilal Ali, Syed Talal Wasim, and Umair Bin Mansoor. Analysing the robustness of vision-language-models to common corruptions. *arXiv preprint arXiv:2504.13690*, 2025. 1, 2

[33] Pablo Valle, Sergio Segura, Shaukat Ali, and Aitor Arrieta. Metamorphic testing of vision-language action-enabled robots. *arXiv preprint arXiv:2602.22579*, 2026. 2

[34] Vincent Walsh and Janusz Kulikowski. *Perceptual Constancy: Why Things Look as They Do*. Cambridge University Press, 2014. 1, 2, 3

[35] Dequan Wang, Evan Shelhamer, Shaoteng Liu, Bruno Olshausen, and Trevor Darrell. Tent: Fully test-time adaptation by entropy minimization. In *International Conference on Learning Representations (ICLR)*, 2021. 2

[36] Peng Wang, Shuai Bai, Sinan Tan, et al. Qwen2-vl: Enhancing vision-language model's perception of the world at any resolution. *arXiv preprint arXiv:2409.12191*, 2024. 1, 3

[37] Shukang Yin, Chaoyou Fu, Sirui Zhao, et al. A survey on multimodal large language models. *National Science Review*, 2024. 1

[38] Hugo Laurençon Zeng, Léo Tronchon, Matthieu Cord, and Victor Sanh. What matters when building vision-language models? *arXiv preprint arXiv:2405.02246*, 2024. 2

[39] Wanqi Zhou, Shuanghao Bai, Danilo P. Mandic, Qibin Zhao, and Badong Chen. Revisiting the adversarial robustness of vision language models: a multimodal perspective. *arXiv preprint arXiv:2404.19287*, 2024. 2